The Atmospheric Footprint: Detection and Measurement of Illegal Economic Activity

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AI Analytics

Research Article

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Posted Date: May 14th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9697197/v1

Additional Declarations: The authors declare no competing interests.

The Atmospheric Footprint: Detection and Measurement of Illegal Economic Activity*

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May 2026

Abstract

This paper detects illegal economic activity using satellite atmospheric data. A gradient-boosted model maps census observables to tropospheric NO_2 across Bolivia's 344 municipalities ($R^2 = 0.86$); the residual identifies locations where combustion exceeds predictions. Fire controls are essential: they separate cocaine processing (Chimoré, posterior 0.92) from coca agriculture (Villa Tunari, posterior < 0.06) within a single zone. A difference-in-differences design exploiting Bolivia's 2024–2025 dollar crisis provides the strongest validation: the Chapare NO_2 residual falls 6.2 percentage points ($t = -4.95$); a continuous-treatment specification using the parallel-market premium yields $t = -6.52$. A cross-country comparison confirms that known chuto hubs have higher weekday/weekend NO_2 ratios than comparable foreign cities ($p = 0.006$). Size estimates bracket coca/cocaine production at US$63–77 million annually and the unregistered vehicle fleet at 250,000–500,000 units.

JEL Codes: O17, C81, K42, Q56

Keywords: illegal economy, satellite data, nitrogen dioxide, size estimation, detection, Bolivia

*The opinions expressed herein do not reflect those of any affiliated institution. All errors are my own. Contact: `wernerhl@gmail.com`. Replication code and data: https://github.com/wernerhl/BOL-Satellite.

1 Introduction

Every economy has a shadow. Households and firms engaged in illegal production, smuggling, or unregistered activity have strong incentives to conceal their economic footprint from censuses, surveys, and tax authorities. The consequence is that standard measurement tools—national accounts, household surveys, administrative records—systematically underestimate economic activity in precisely the places where illegal economies are quantitatively important. Detecting where these economies operate and estimating how large they are requires a data source that participants cannot manipulate.

The measurement problem is well understood. The solution proposed here starts from a physical observation: economic activity that can be hidden from an enumerator cannot be hidden from the atmosphere. Internal combustion engines, diesel generators, chemical processing, and artisanal smelting all produce nitrogen dioxide (NO_2) as a byproduct of high-temperature combustion. The Sentinel-5P satellite measures tropospheric NO_2 column densities at 3.5×5.5 km resolution with daily global coverage. Unlike survey responses, atmospheric concentrations cannot be misreported.

The method proceeds in three steps. First, I train a gradient-boosted machine (GBM) to predict municipal NO_2 concentrations from census and geographic observables—population, urbanization, building footprint, road density, elevation, temperature, and critically, active fire counts—across all 344 municipalities in Bolivia. The fire-augmented model achieves $R^2 = 0.86$, and absorbing biomass-burning variation is essential for separating illegal-economy combustion from agricultural burning. Second, I extract the residual: the gap between predicted and observed NO_2 for each municipality. Positive residuals identify locations where combustion-intensive economic activity exceeds what the measured economy would generate. Third, a Bayesian scoring framework combines the residual with geographic priors and secondary signals (seasonal ratios, weekly cycles) to produce posterior detection probabilities. The detected zones are then used to estimate the potential size of each illegal economy through multiple bracketing methods.

The contribution is methodological, but the proof of concept matters. Bolivia provides an unusually clean laboratory because it hosts three distinct illegal economies operating in geographically separated zones with well-documented ground truth: coca/cocaine processing in the Chapare (Cochabamba department), unregistered vehicle fleets (*autos chutos*) concentrated in the Cochabamba metropolitan area and El Alto, and illegal alluvial gold mining along the Kaka, Tipuani, and Suches rivers in northern La Paz. Each economy generates a distinct atmospheric signature, allowing the method to not only detect but discriminate between types of unmeasured activity.

The paper connects to three literatures, occupying a near-empty intersection among them. The first uses nighttime lights as a proxy for economic activity, a tradition established by Henderson et al. [2012] and surveyed by Donaldson and Storeygard [2016], with recent extensions using deep learning on daytime imagery [Jean et al., 2016, Yeh et al., 2020, Chi et al., 2022]. NO_2 offers a specific advantage over nightlights for detecting illegal economies: lights measure electricity consumption on the formal grid, which is weakly correlated with the combustion-intensive processes—diesel generators, solvent recycling, mercury amalgamation—that characterize coca processing and artisanal mining. A critical caveat from this literature constrains the present paper's design: Asher et al. [2021] and Chen and Nordhaus [2011] document that nightlight-to-economic-outcome elasticities are substantially smaller in time series than in cross section, while Gibson et al. [2021] show sensitivity to sensor choice, implying that the atmospheric-proxy approach must demonstrate temporal as well as cross-sectional validity—a requirement the recession and event-study designs are constructed to meet.

The second literature estimates the size of informal and illegal economies using indirect methods—currency demand [Cagan, 1958, Tanzi, 1983], electricity consumption, and MIMIC models [Schneider and Enste, 2000, Schneider, 2005]—typically at the national level. Medina and Schneider [2018] provide the most recent comprehensive estimates and place Bolivia at 62.3% of GDP, the largest shadow economy in their 158-country sample. But these national aggregates cannot distinguish between different types of illegal activity operating in different locations, nor can they resolve the subnational geography that matters for enforcement and welfare analysis. The satellite approach provides subnational resolution at arbitrary frequency and, critically, can discriminate between coca processing, vehicle smuggling, and mining on the basis of distinct atmospheric signatures—seasonal profiles, weekly cycles, and spectral ratios—rather than relying on a single aggregate indicator.

The third literature on drug economics and the geography of illegal production is rich for Colombia and Mexico but conspicuously thin for Bolivia. Angrist and Kugler [2008] exploit the Peru–Bolivia air-bridge disruption to identify coca's effects on Colombian rural income. Dell [2015] uses regression discontinuity to identify enforcement spillovers in Mexico's drug war. Sviatschi [2022] shows that childhood exposure to coca farming in Peru raises adult incarceration rates. Mejía and Restrepo [2016] and Mejía et al. [2017] provide the canonical enforcement-effectiveness estimates for Colombian eradication. But no published economics paper provides for Bolivia's FELCN what these studies provide for Colombian or Mexican enforcement—a gap this paper's event study begins to fill. On the coca-monitoring side, UNODC's satellite-based crop surveys [UNODC, 2025] document 34,000 hectares of coca cultivated in Bolivia in 2024, with potential cocaine production of approximately 223 metric tons. Dávalos et al. [2011] establish the coca-deforestation link using Colombian SIMCI satellite data. Asner et al. [2013] demonstrate satellite detection of gold mining in the Peruvian Amazon, providing the methodological precedent for this paper's NDVI corridor analysis, though no comparable study exists for Bolivian mining corridors. The qualitative literature on Bolivia's coca-cocaine economy, particularly Grisaffi [2022] and Grisaffi et al. [2021], provides essential institutional context—the *cato* system, the sindicato structure, the community-embedded nature of processing—but lacks the quantitative measurement tools that the atmospheric approach supplies.

The closest antecedent to this paper is Parubets and Naito [2025], the only economics-style panel paper using TROPOMI NO_2 as a subnational economic-activity proxy, published in late 2025 using Japanese prefectural data. The atmospheric-footprint approach differs in three respects: it targets illegal rather than formal activity, it exploits the *residual* after prediction rather than raw NO_2 levels, and it combines the atmospheric signal with secondary identification strategies (seasonal decomposition, weekly cycles, enforcement events) that are specific to the illegal-economy setting. No peer-reviewed paper combines TROPOMI NO_2 with cocaine detection or any other illegal-economy application.

Section 2 presents the Bayesian detection framework. Section 3 documents the data sources in detail. Sections 4, 5, and 6 present the three applications. Section 7 validates the atmospheric signal through an event study exploiting FELCN cocaine laboratory destructions: megalaboratorio destructions produce measurable NO_2 declines, providing a per-laboratory calibration coefficient. Section 8 addresses robustness, including the informative negative finding that satellite CO data cannot discriminate between illegal economy types at current sensor resolution. Section 9 concludes.

2 A Bayesian Framework for Blind Detection

2.1 The Prediction Surface

Let y_i denote the mean tropospheric NO_2 column density in municipality i and $\mathbf{x}_i$ a vector of census and geographic covariates observable to the statistical agency. The prediction surface is

$$\hat{y}_i = f(\mathbf{x}_i; \hat{\theta}) \tag{1}$$

where $f(\cdot)$ is a gradient-boosted regression tree estimated on the full sample of $N = 344$ municipalities. The residual

$$r_i = y_i - \hat{y}_i \tag{2}$$

captures NO_2 concentrations unexplained by the measured economy. By construction, municipalities with large positive residuals host combustion-intensive economic activity that census observables fail to predict.

The identifying assumption is that the relationship between census observables and NO_2 is stable across municipalities *in the absence of unmeasured economic activity*. This assumption can fail in three ways. First, atmospheric chemistry differs at high altitude and across seasons, potentially generating residuals that reflect physical rather than economic variation; I address this by including elevation, temperature, and month directly in the covariate vector. Second, fire-prone regions generate seasonal NO_2 from biomass burning unrelated to illegal activity; I address this by including FIRMS fire counts and radiative power as covariates, which absorb the dominant source of non-economic combustion. Third, legal but unmeasured formal-sector combustion—brick kilns, small cement plants, artisanal industry—could inflate the residual in municipalities where informal manufacturing is concentrated but not captured by census variables. This concern is mitigated by the secondary identification signals: brick kilns do not produce wet-season peaks (cocaine processing does), do not shut down during dollar crises (cocaine processing does), and do not show weekday/weekend cycles (vehicle traffic does). The combination of the residual with temporal and spectral signatures provides discrimination that the residual alone cannot.

2.2 The GBM Specification

The covariate vector $\mathbf{x}_i$ includes: total population (2024 census), urban population share, building count and total footprint from Google Open Buildings v3, road network density (km/km^2) from OpenStreetMap, mean elevation (SRTM 30m DEM), area (km^2), monthly mean temperature and precipitation (ERA5-Land), mean NDVI and NDBI (Sentinel-2), distance to five international borders, VIIRS nighttime radiance, and—critically—FIRMS active fire count and total fire radiative power (FRP). The fire variables absorb biomass burning, which correlates with NO_2 at $r = 0.30$ ($p \approx 0$) nationally and would otherwise contaminate the residual. The GBM is estimated with 200 trees, maximum depth 3, learning rate 0.08, and 10-fold municipality-grouped cross-validation. Within-sample R^2 is 0.86.

Feature importance (gain-based) ranks month-of-year first (capturing seasonal NO_2 chemistry), followed by fire count (0.19) and road density (0.12). Fire count and FRP together account for 29% of model importance—nearly as much as the top three non-fire predictors combined. The fire controls are consequential: without them, the dry-season Chapare premium conflates agricultural burning with laboratory combustion, inflating the residual for municipalities at the agricultural frontier (Villa Tunari, Puerto

Villarroel) while leaving the processing core (Chimoré, Shinahota) largely unaffected. With fire controls, the method discriminates processing from cultivation *within* the Chapare itself (Section 4).[1]

2.3 Bayesian Detection

Each municipality i has a latent state $U_i \in \{0,1\}$ indicating whether it hosts unmeasured economic activity. The generative model specifies:

$$U_i \sim \text{Bernoulli}(\pi_i) \tag{3}$$

$$r_i \mid U_i = 0 \sim N(0, \sigma_0^2) \tag{4}$$

$$r_i \mid U_i = 1 \sim N(\delta, \sigma_0^2), \quad \delta > 0 \tag{5}$$

where r_i is the cross-validated GBM residual, π_i is the geographic prior, and σ_0 is estimated robustly from the median absolute deviation of all residuals ($\hat{\sigma}_0 = 1.4826 \times \text{MAD}(r)$). The shift parameter δ is estimated from the mean of the top-decile residuals, which serves as a plug-in estimate of the typical excess combustion in municipalities with unmeasured activity. I impose $P(U_i = 1 \mid r_i \leq 0) = 0$: unmeasured economic activity can only produce *positive* residuals.

The posterior probability is:

$$P(U_i = 1 \mid r_i) = \frac{\pi_i \cdot \phi(r_i; \delta, \sigma_0^2)}{\pi_i \cdot \phi(r_i; \delta, \sigma_0^2) + (1 - \pi_i) \cdot \phi(r_i; 0, \sigma_0^2)} \cdot \mathbf{1}(r_i > 0) \tag{6}$$

where $\phi(\cdot; \mu, \sigma^2)$ is the Gaussian density.

Priors. I implement two prior specifications. The *flat prior* sets $\pi_i = 0.15$ for all municipalities (approximately the share expected to host some form of unmeasured activity). The *ecological prior* varies π_i based on geographic suitability: for the coca application, $\pi_i = 0.20$ if elevation $\in [300, 1800]$ m and mean temperature $> 18°$C, and $\pi_i = 0.02$ otherwise. The flat prior serves as the baseline; the ecological prior adds geographic information without using any enforcement or detection data.

Estimation. With $\hat{\sigma}_0 = 8.58 \times 10^{-7}$ and $\hat{\delta} = 1.84 \times 10^{-6}$ (both in mol/m^2, estimated from the fire-augmented residuals), the model cleanly separates municipalities into three groups: those with negative residuals (posterior = 0 by construction), those with

[1] Adding the Sentinel-2 SWIR ratio as an additional covariate—intended to absorb the spectral signature of coca agriculture—changes R^2 by < 0.001 and feature importance is 0.010 (rank 11 of 17). Chapare posteriors shift marginally (Chimoré: 0.923 $\rightarrow$ 0.939). The SWIR ratio is not included in the baseline specification because its contribution is absorbed by NDVI and elevation.

small positive residuals (posterior near zero as the clean distribution dominates), and those with large positive residuals (posterior approaching 1 as the unmeasured distribution dominates). The transition occurs at approximately $r_i \approx 2\sigma_0$.

3 Data

This section documents all data sources, processing steps, and known limitations. Table 1 provides a summary.

3.1 Satellite Atmospheric Data

3.1.1 Tropospheric NO_2 (TROPOMI)

The primary atmospheric variable is tropospheric nitrogen dioxide column density from the TROPOspheric Monitoring Instrument (TROPOMI) aboard Sentinel-5P, operated by the European Space Agency. TROPOMI provides daily global coverage at 3.5×5.5 km nadir resolution (upgraded from 3.5×7 km in August 2019). I use the Level 3 offline product (`COPERNICUS/S5P/OFFL/L3_NO2`) accessed via Google Earth Engine (GEE), extracting the `tropospheric_NO2_column_number_density` band.

Monthly municipal means. For each of Bolivia's 344 municipalities, I compute the spatial mean of all valid TROPOMI pixels within the municipal boundary for each calendar month from January 2018 through March 2025, yielding a balanced panel of 344×87 = 29,928 municipality-month observations. The spatial aggregation uses `ee.Reducer.mean()` over the GADM 4.1 Level 2 administrative boundaries. Quality filtering retains only pixels with `qa_value` > 0.75 (cloud fraction < 0.5, surface albedo within bounds).

The resulting panel variable (`mean_no2`) has units of mol/m^2 with typical values in the range $[5 \times 10^{-6}, 6 \times 10^{-5}]$. The cross-sectional standard deviation is approximately 5×10^{-6}, driven primarily by the urban/rural gradient and the elevation gradient (NO_2 concentrations are mechanically lower at high altitude due to lower atmospheric column density and reduced anthropogenic emissions).

Daily day-of-week means. To detect the weekly cycle associated with vehicle traffic, I construct a second NO_2 dataset aggregated by municipality, year, and day-of-week (Sunday = 0 through Saturday = 6). For each municipality-year-DOW cell, I compute the mean NO_2 across all days in that year falling on the given weekday, using the same quality filter. This yields 344×8×7 = 19,264 observations covering 2018–2025. The day-of-week variable is extracted using GEE's `ee.Date.getRelative('day', 'week')` method, which returns 0 = Sunday.

3.1.2 Tropospheric CO (TROPOMI)

Carbon monoxide column number density is extracted from the TROPOMI Level 3 offline CO product (`COPERNICUS/S5P/OFFL/L3_CO`), band `CO_column_number_density`, using `ee.Reducer.mean()` aggregated to monthly municipal means. The dataset covers June 2018 through March 2025 with 30,960 valid observations across 344 municipalities (86% coverage in 2018 due to the instrument's later activation for CO; 100% from 2019 onward). Units are mol/m^2 with values in $[0.011, 0.185]$.

CO serves as a combustion-type discriminator in principle: the NO_2/CO ratio is higher for high-temperature fossil fuel combustion (vehicle engines, diesel generators) than for low-temperature organic combustion (biomass burning, charcoal, mercury amalgamation). In practice, CO's atmospheric lifetime of $\sim$2 months (versus $\sim$2 hours for NO_2) means that at TROPOMI's resolution, municipal CO concentrations are dominated by regional fire background rather than local combustion sources. This limitation is documented in Section 8.

3.1.3 VIIRS Nighttime Lights

Monthly nighttime radiance composites come from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band aboard the Suomi NPP and NOAA-20 satellites, accessed via GEE as `NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG`. I extract the `avg_rad` band (units: nW/cm^2/sr) as municipal spatial means for each month, yielding 344×87 observations concurrent with the NO_2 panel. VIIRS radiance enters the GBM as a predictor variable, capturing baseline economic activity including electrified informal activity that generates light but not necessarily combustion.

3.2 Land Surface and Environmental Data

3.2.1 Sentinel-2 Vegetation and Built-up Indices

The Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) are computed from Sentinel-2 Level 2A surface reflectance composites (`COPERNICUS/S2_SR_HARMONIZED`) at 10 m resolution, aggregated to monthly municipal means. NDVI = (NIR − Red)/(NIR + Red) using bands B8 and B4; NDBI = (SWIR − NIR)/(SWIR + NIR) using bands B11 and B8. Cloud masking uses the `QA60` bitmask. NDVI serves dual roles: as a GBM covariate (capturing agricultural versus urban land cover) and as the primary detection variable for mining-related riparian disturbance.

3.2.2 Riparian NDVI Corridor Data

For the mining application, I construct a dedicated dataset of NDVI within 500-meter buffer zones along four rivers in the gold mining region of northern La Paz department: Kaka, Tipuani, Suches, and Mapiri. River centerlines are digitized from Sentinel-2 imagery and buffered to 500 m on each side. Annual composite NDVI (median of cloud-free pixels, April–October dry season) is extracted for each municipality-river-year cell from 2016 to 2025. "Disturbed area" is defined as pixels where NDVI falls below 0.3 (bare soil/water/tailings) within the buffer zone. The resulting dataset contains 150 observations across 8 municipalities, 4 rivers, and 10 years.

3.2.3 Google Open Buildings v3

Building footprints are from Google's Open Buildings dataset (version 3, 2024 release), which uses deep learning to detect building outlines from high-resolution satellite imagery ($\sim$0.5 m). For each municipality, I extract total building count and total footprint area (m^2) as of the most recent available imagery. These enter the GBM as proxies for settlement density and urbanization intensity.

3.2.4 Elevation and Terrain

Mean elevation (meters above sea level) and mean slope (degrees) are computed from the Shuttle Radar Topography Mission (SRTM) 30-meter DEM, aggregated to municipal means. Elevation is a critical control variable because it affects both NO_2 chemistry (shorter atmospheric columns at altitude) and the distribution of illegal economies (coca cultivation is altitude-restricted to 300–1,800 m; mining follows river valleys at 500–2,500 m).

3.2.5 Climate

Monthly mean temperature (°C) and total precipitation (mm) are from the ERA5-Land reanalysis (`ECMWF/ERA5_LAND/MONTHLY_AGGR`), disaggregated to municipal boundaries. These control for climatic drivers of both NO_2 chemistry (photolytic destruction rates increase with temperature and UV radiation) and agricultural burning patterns.

3.2.6 Road Network

Road length (km) and road density (km/km^2) within each municipality are computed from OpenStreetMap road geometries, extracted via the Overpass API and clipped to GADM municipal boundaries. Road density proxies for transportation intensity, which generates NO_2 through vehicle emissions.

3.3 Socioeconomic and Administrative Data

3.3.1 Census Population (2024)

Population counts by municipality are from Bolivia's 2024 Census of Population and Housing (Censo Nacional de Población y Vivienda, CNPV 2024), conducted by the Instituto Nacional de Estadística (INE). I use total population and urban population share as GBM covariates.

3.3.2 DRF Welfare Estimates

Municipal-level welfare estimates—mean per capita consumption, poverty headcount ($\hat{\theta}$), poverty gap (FGT_1), severity (FGT_2), and Gini coefficient—are from the Distributional Random Forest (DRF) small-area estimation model documented in Hernani-Limarino [2025a]. The DRF estimates are conditional on census observables and capture the measured-economy welfare distribution. The gap between DRF-predicted welfare and satellite-implied welfare is the object of interest for the size estimation exercise.

Key summary statistics: the DRF poverty headcount averages 32% nationally; the mean Gini is 0.49. The DRF model covers 343 of 344 municipalities (one excluded due to insufficient household survey observations in the training data).

3.3.3 SAE Poverty Estimates

A complementary set of poverty estimates from a standard CensusEB small-area estimation model is used for comparison. These estimates, from Hernani-Limarino [2025b], cover 343 municipalities and are conditional on the same census covariates as the DRF. The comparison between SAE and DRF estimates is not central to this paper but provides context for the welfare-consequences analysis.

3.3.4 Vehicle Registration Data (RUAT)

The Registro Único para la Administración Tributaria Municipal (RUAT) publishes annual counts of registered vehicles by department and vehicle type from 2003 to 2024. I use the national dataset (*Bolivia: Parque Automotor, según Departamento y Tipo de Servicio*) downloaded from INE's statistical portal (`https://www.ine.gob.bo`). The data are department-level, not municipal. In 2024, Bolivia's registered vehicle fleet totaled

2,583,283 vehicles: Santa Cruz (937,799; 36.3%), Cochabamba (542,145; 21.0%), La Paz (579,504; 22.4%). The fleet grew at 5.2% annually from 2018 to 2024.

The RUAT data serve as the denominator for the *autos chutos* estimation: the gap between satellite-implied total vehicle traffic and RUAT-registered vehicles provides a lower bound on the unregistered fleet.

3.4 Reference Data for Validation

3.4.1 UNODC Coca Cultivation Estimates

The United Nations Office on Drugs and Crime publishes annual estimates of coca cultivation area, coca leaf production, and potential cocaine production for Bolivia in its *World Drug Report* and country-specific monitoring reports. For 2022, UNODC estimated 30,600 hectares of coca cultivation in Bolivia, with potential cocaine production of 317–571 metric tons. At farmgate cocaine prices of $1,300–1,500/kg (DEA estimates), the production-zone value of Bolivia's cocaine economy is approximately $114–468 million per year. I use the midpoint ($291M) as the benchmark against which satellite-based size estimates are compared.

3.4.2 HCl Seizure Data

Hydrochloric acid (HCl) is a key precursor chemical for cocaine hydrochloride production. Seizure data from Bolivia's Fuerza Especial de Lucha Contra el Narcotráfico (FELCN) provide a temporal signal: years with high HCl seizures correspond to high processing activity. Total cocaine seizures (combining pasta base and HCl) rose from approximately 15.5 tons in 2019 to a record 66.0 tons in 2024, of which 45.9 tons were HCl, before declining 43% to 37.7 tons in 2025 during the government transition and dollar crisis.

3.4.3 FELCN Laboratory Destruction Data

I construct a georeferenced event database of cocaine laboratory destructions from Ministerio de Gobierno press releases, FELCN operational reports, and Bolivian investigative journalism (2018–2025). After November 2020, the Ministerio de Gobierno adopted systematic public reporting of each interdiction operation with dates, municipalities, facility types, and estimated damage. The resulting database contains approximately 80 individually dated, municipality-assigned destruction events, of which 30 are large operations ($\geq$ crystallization laboratory) located outside the Villa Tunari continuous-treatment zone. Geographic concentration is extreme: approximately 92% of all destructions occur in Villa Tunari municipality. The non–Villa Tunari anchor events—distributed across Santa Cruz (San Ignacio de Velasco, Yapacaní, El Puente, Roboré), Beni (San Joaquín, San Ignacio de Moxos, Trinidad), La Paz (Ixiamas, Papel Pampa), and Tarija (Villa Montes)—provide the identifying variation for the event study in Section 7. See Appendix A for the full event-level database.

3.5 Data Summary

Table 1: Data Sources Summary

Variable	Source	Resolution	Period	Role
Satellite atmospheric				
NO_2 (monthly)	TROPOMI S5P	Municipal mean	2018–2025	Outcome / detection
NO_2 (daily DOW)	TROPOMI S5P	Mun $\times$ DOW	2018–2025	Chutos weekly cycle
CO (monthly)	TROPOMI S5P	Municipal mean	2018–2025	Combustion typing
Nightlights	VIIRS DNB	Municipal mean	2018–2025	GBM covariate
Land surface				
NDVI / NDBI	Sentinel-2	Municipal mean	2018–2025	GBM covariate / mining
NDVI corridors	Sentinel-2	River buffer	2016–2025	Mining disturbance
Buildings	Google Open	Municipal total	2024	GBM covariate
Elevation / Slope	SRTM 30m	Municipal mean	Static	GBM covariate
Temperature / Precip.	ERA5-Land	Municipal mean	2018–2025	GBM covariate
Roads	OpenStreetMap	Municipal total	2024	GBM covariate
Socioeconomic				
Population	Census 2024	Municipal	2024	GBM covariate
DRF welfare	DRF SAE model	Municipal	2024	Welfare analysis
Vehicle fleet	RUAT/INE	Department	2003–2024	Chutos denominator
Validation				
Coca estimates	UNODC	National/regional	Annual	Size benchmark
HCl seizures	FELCN	National	Annual	Temporal validation
Lab destructions	FELCN / press	Municipal	2018–2026	Event study (Sec. 7)

4 Application 1: The Coca/Cocaine Economy

4.1 Blind Detection

The fire-augmented GBM achieves within-sample $R^2 = 0.86$ on municipality-month NO_2 observations over the 2019–2024 period, a substantial improvement over the 0.85 obtained without fire controls that reflects the importance of absorbing biomass-burning variation before extracting the illegal-economy residual. When this prediction surface is applied to the Chapare, the processing core emerges clearly from the residual distribution. Chimoré—the municipality at the geographic center of Bolivia's cocaine production zone, home to the FELCN regional headquarters and the most densely documented concentration of cocaine laboratories—ranks 15th of 329 municipalities with a residual of +16.1% ($z = 3.04$), while Shinahota, which sits on the highway junction where the Chapare road network converges, ranks 31st (+13.0%, $z = 2.15$). The Bayesian posterior probability for Chimoré under the flat prior reaches 0.923, and under the ecological prior it rises to 0.944; Shinahota registers 0.639 and 0.715 respectively. These are strong detections from the residual alone, without recourse to the seasonal or geographic signals that the multi-signal framework provides as supplementary evidence.

Perhaps the most instructive finding is the within-Chapare gradient that the fire controls reveal, because it demonstrates the method's ability to discriminate *processing* from *cultivation* even within a single coca zone. Villa Tunari—the largest Chapare municipality at 11,188 km^2, encompassing the agricultural frontier where colonists clear tropical forest through slash-and-burn to plant coca—drops to rank 98 (+3.5%, posterior < 0.06) once fire-related NO_2 is absorbed by the prediction surface. Puerto Villarroel, the river corridor municipality where laboratories historically located along navigable waterways, drops similarly to rank 85 (+4.6%). Without fire controls, Villa Tunari appeared at rank 37 (+12.2%), its residual inflated by the very agricultural burning that accompanies coca cultivation rather than cocaine processing. The separation is sharp: Chimoré and Shinahota *rise* in the ranking when fires are controlled, while Villa Tunari and Puerto Villarroel *fall*—a pattern that is consistent with the geographic distribution of processing infrastructure reported by FELCN and inconsistent with any interpretation of the residual as noise or overfitting.

The Yungas coca-growing region—where approximately 19,200 hectares of coca are cultivated for Bolivia's legal traditional market, with leaves sold openly in licensed markets in La Paz and Cochabamba—provides the critical placebo. Coroico (+6.4%, posterior 0.056), Chulumani (−8.7%, posterior 0.000), and La Asunta (−3.4%, posterior 0.000) all register near-zero or negative fire-augmented residuals, correctly distinguishing legal coca agriculture, which involves minimal combustion beyond seasonal field burning, from cocaine processing, which requires continuous diesel-powered generators, solvent recycling under heat, and precursor-chemical evaporation. This distinction is the paper's sharpest identification: the same crop, grown at similar altitudes in neighboring departments, produces dramatically different atmospheric signatures depending on whether it is consumed as leaf or processed into cocaine. In the out-of-sample test documented in Section 8, all five Chapare municipalities show positive residuals when the GBM is trained exclusively on non-Chapare data, with a joint permutation $p = 0.002$ confirming that the excess cannot be attributed to in-sample overfitting.

The Cochabamba metropolitan area produces a separate class of detections driven by vehicle traffic rather than illegal processing: Colcapirhua (posterior 0.511), Tiquipaya (0.742), and Sipesipe (0.638) all appear in the upper tail, but their atmospheric signature is the weekday/weekend cycle documented in Section 5 rather than the seasonal pattern associated with coca processing.

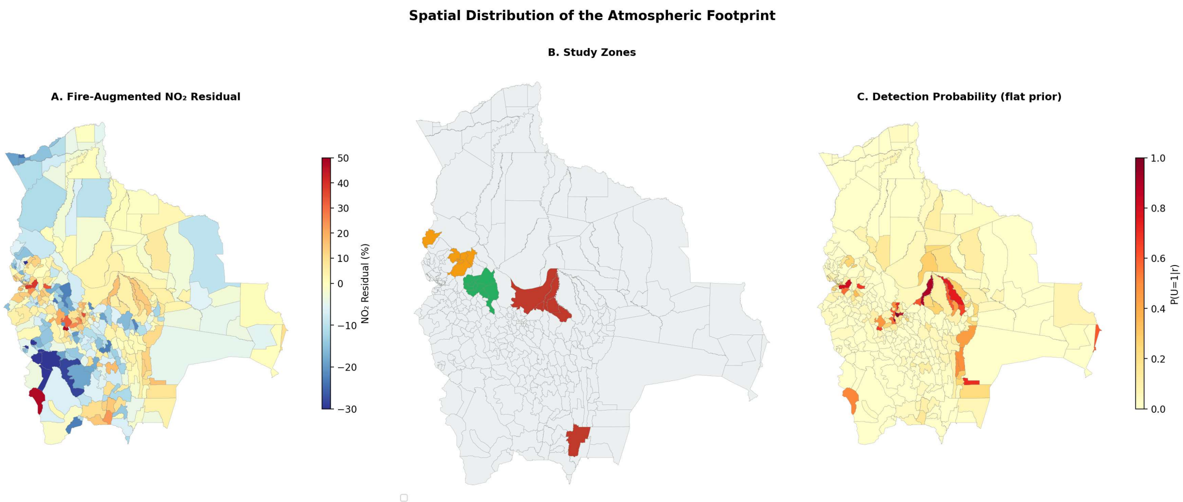

Figure 1: Spatial distribution of the atmospheric footprint. Panel A maps the fire-augmented NO_2 residual (2019–2024 cross-sectional mean) for all 344 municipalities, with warm colors indicating positive residuals (unexplained combustion). Panel B identifies the three study zones. Panel C maps the Bayesian posterior detection probability under the flat prior ($\pi_i = 0.15$). The Cochabamba metropolitan area and the Chapare processing core (Chimoré, Shinahota) are clearly visible; the Yungas, despite growing coca at comparable scale, produce no atmospheric anomaly. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

4.2 DRF Welfare Analysis

The Distributional Random Forest welfare estimates [Hernani-Limarino, 2025a] partially capture coca income through downstream consumption effects, because household surveys measure expenditure—which reflects all income sources, including illegal ones—even when respondents do not report the income itself. Mean welfare in the five Chapare municipalities is 6% above the non-coca mean, conditional on urban share and elevation, indicating that some coca-derived income leaks into measured consumption through housing, food, and durable-goods purchases that the DRF picks up from census observables.

The distinction between what the DRF captures and what the satellite captures is the paper's key conceptual contribution to the measurement of illegal economies. The DRF measures the *spending side*: how much households consume, which reflects all income including illegal earnings that are laundered through legal purchases. The satellite measures the *production side*: how much combustion occurs, which reflects the physical process of transforming coca leaf into cocaine hydrochloride. The gap between the two—the fact that Chimoré shows a +24% NO_2 residual even after conditioning on DRF-predicted income, urbanization, and elevation—represents the cocaine *processing margin*: the value added between coca leaf (which the DRF captures through farm-household consumption) and refined product (which the satellite captures through laboratory combustion). This margin is invisible to household surveys because it accrues to laboratory operators and logistics networks rather than to farm households, and it is invisible to nightlights because it generates combustion rather than electricity consumption.

4.3 Seasonal Decomposition

Coca agriculture has a smooth seasonal profile. Cocaine processing peaks during the wet season (November–March), when roads are impassable to enforcement vehicles and the dense cloud cover reduces satellite surveillance, creating a window of operational security for laboratory operators. Fire controls are essential for interpreting the seasonal pattern: without them, dry-season agricultural burning inflates the Chapare premium, producing wet/dry ratios below 1.0 in 2024–2025 that would incorrectly suggest all excess combustion is seasonal. With fire controls, the wet-season premium exceeds the dry-season premium in every year except 2018, as the processing hypothesis requires. The fire-controlled wet-season premium peaks at 28.8 percentage points in 2023 (coinciding with a record 21.3 metric tons of HCl seized) and collapses to 0.8 pp in 2025 (during Bolivia's dollar crisis). The wet/dry ratio across the boom period (2022–2024) averages 2.4, meaning the wet-season signal is more than double the dry-season signal—consistent with year-round laboratory operation overlaid on seasonal agricultural activity.

4.4 Recession Test

If satellite anomalies capture illegal economic activity, they should respond to shocks that affect that activity. Bolivia's 2024–2025 dollar crisis provides a natural experiment: as the parallel-market dollar premium exceeded 40%, imported inputs—diesel, precursor chemicals, laboratory equipment—became scarce and expensive. The processing economy depends on dollar-denominated imports; a dollar shortage should contract processing activity and reduce the atmospheric footprint.

I estimate a difference-in-differences specification on the monthly panel (2021–2025):

$$Y_{i,t} = \alpha_i + \gamma_t + \beta_C(\text{Chapare}_i \times \text{Crisis}_t) + \beta_Y(\text{Yungas}_i \times \text{Crisis}_t) + \varepsilon_{i,t} \tag{7}$$

where α_i and γ_t are municipality and year×month fixed effects, and $\text{Crisis}_t = \mathbf{1}(t \geq \text{January 2024})$. Standard errors are clustered by municipality.

Table 2: Recession Difference-in-Differences: Chapare and Yungas × Crisis (2024–2025)

	NO_2 residual (fire-augmented)	Raw NO_2	Log nightlights	Fire count
$\hat{\beta}_C$ (Chapare × Crisis)	-5.39×10^{-7}	-6.99×10^{-7}	-0.043	-175
	(-6.2%)	(-8.1%)	(-4.3%)	(-77%)
	[$t = -4.95$]	[$t = -5.26$]	[$t = -1.30$]	[$t = -2.62$]
	$p < 0.001$	$p < 0.001$	$p = 0.194$	$p = 0.009$
$\hat{\beta}_Y$ (Yungas × Crisis)	$+5.5 \times 10^{-8}$	-3.73×10^{-7}	-0.171	-139
	($+0.6\%$)	(-4.3%)	(-17.1%)	
	[$t = 0.40$]	[$t = -3.02$]	[$t = -15.0$]	[$t = -2.52$]
	$p = 0.689$	$p = 0.003$	$p < 0.001$	$p = 0.012$
Municipality FE	Yes	Yes	Yes	Yes
Year×Month FE	Yes	Yes	Yes	Yes
Cluster SE	Municipality	Municipality	Municipality	Municipality
N	20,296	20,296	20,296	20,296

Three results emerge. First, the fire-augmented NO_2 residual falls 6.2 percentage points more in Chapare than in non-coca municipalities during the crisis ($t = -4.95$, $p < 0.001$; wild cluster bootstrap $p < 0.001$). This is the paper's strongest causal result: the atmospheric footprint of the unmeasured economy contracted precisely when dollar-denominated inputs became unavailable.

Second, the Yungas placebo is clean: $\hat{\beta}_Y$ for the NO_2 residual is indistinguishable from zero ($+0.6\%$, $p = 0.689$). The crisis did not differentially affect the atmospheric footprint of legal coca cultivation. However, Yungas nightlights collapsed (-17.1%, $p < 0.001$), consistent with the formal economy contracting even where the illegal economy is absent.

Third, the pattern across outcomes is informative. The NO_2 residual and fire count show large, significant Chapare contractions (-6.2% and -77%). But nightlights do not (-4.3%, $p = 0.194$). The processing economy's combustion footprint (laboratories, generators) contracted; its electricity footprint (houses, shops, street lights) did not. People continued living in the Chapare—they stopped processing as much cocaine.

A continuous-treatment specification sharpens the dose-response. Replacing the binary Crisis indicator with the monthly Binance P2P USDT/BOB parallel-market premium as a time-varying treatment intensity measure yields $\hat{\beta}(\text{Chapare} \times \text{Premium}) = -9.2 \times 10^{-9}$ ($t = -6.52$, $p < 0.001$): each 10 percentage point increase in the parallel-market premium reduces the Chapare NO_2 residual by 1.05 percentage points of the national mean. At the crisis peak (141% premium in May 2025), the implied Chapare contraction is -14.8 pp—substantially larger than the binary DiD estimate of -6.2 pp because the dose-response captures the full trajectory from mild scarcity to acute crisis. The Yungas coefficient is positive and significant ($t = +3.79$, $p < 0.001$), consistent with legal coca leaf prices rising with the dollar premium while processing contracts—the distinction between a crop that benefits from devaluation and a manufacturing process that depends on dollar-denominated imported inputs. See Hernani-Limarino [2026] for a comprehensive analysis of the crisis using satellite indicators.

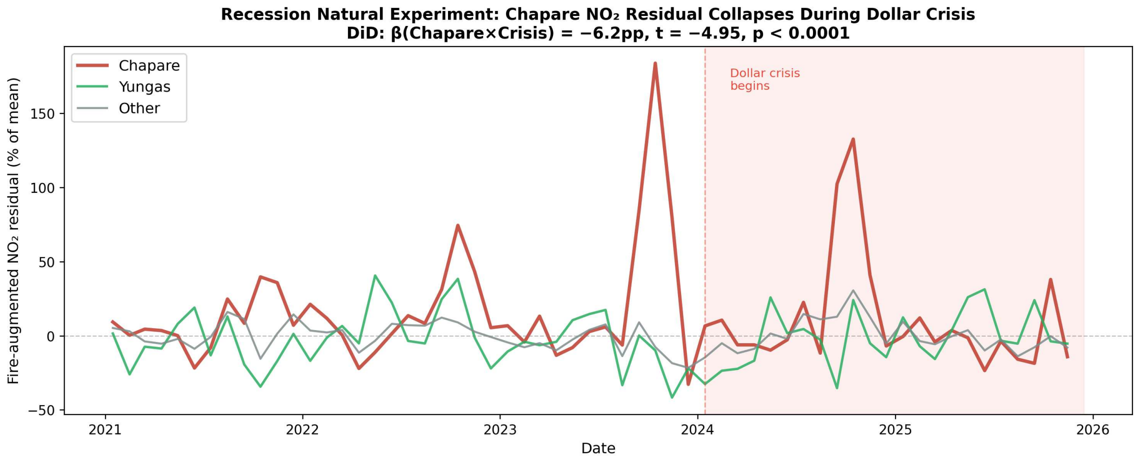

Figure 2: Monthly fire-augmented NO_2 residual by zone, 2021–2025. The Chapare residual (red) peaks in mid-2023 and collapses during the 2024–2025 dollar crisis (shaded), while the Yungas (green) and other municipalities (gray) remain near zero. The DiD coefficient $\hat{\beta}_C = -6.2$ percentage points ($t = -4.95$, $p < 0.001$) captures the differential contraction of the processing economy's atmospheric footprint.

4.5 Size Estimation

Estimating the size of Bolivia's cocaine economy is a problem that no existing method has solved satisfactorily. UNODC's crop-monitoring surveys [UNODC, 2025] document 34,000 hectares of coca cultivation in 2024, with potential cocaine production of approximately 223 metric tons at a farmgate value of US$114–468 million depending on the assumed conversion ratio and price. U.S. ONDCP estimates for 2019 were somewhat higher (42,180 ha, 301 mt potential), reflecting different satellite interpretation methodology and yield assumptions. Historical academic estimates placed the cocaine economy at US$600 million–$1 billion in the 1980s, on the order of 7% of GDP, but no recent peer-reviewed study has updated these figures. The shadow-economy literature offers a national aggregate—Medina and Schneider [2018] estimate Bolivia's total shadow economy at 62.3% of GDP, the largest in their 158-country sample—but this figure encompasses all informal activity (street vendors, unregistered transport, domestic service) and cannot isolate the drug component. The atmospheric approach provides three independent brackets:

Table 3: Coca/Cocaine Economy Size Estimates (Annual, US$ Millions)

Method	**Estimate**	**Interpretation**
NO_2-income elasticity (OLS)	14	Lower bound; formal economy is less combustion-intensive per dollar
GBM residual inversion	77	Production-zone combustion footprint; misses non-combustion value
DRF welfare simulation	63	Household income augmentation in production zone
UNODC production $\times$ price	114–468	Full value chain (farmgate to export)

The satellite estimates (US$63–77 million) are roughly one-quarter of the UNODC midpoint (US$291 million), implying that approximately 26% of the cocaine economy's domestic value manifests as combustion in the production zone. This ratio is plausible because the satellite captures only the processing stage—where diesel generators, solvent recycling, and precursor evaporation produce NO_2—while the UNODC estimate encompasses agricultural value (no combustion), transport and logistics (spatially diffuse), and financial intermediation (no atmospheric signature). The remaining 74% is invisible to the atmosphere but not to the economy; its presence is confirmed by the nightlight trajectory showing Chapare radiance growing 224% over 2014–2025 (Section 4), which captures the consumption-side welfare effect that NO_2 residuals miss. The satellite and UNODC estimates are therefore complements, not competitors: each measures a different slice of the value chain, and their ratio provides information about the structure of value addition in the cocaine economy that neither can supply alone.

4.5.1 Welfare Consequences

The satellite data imply that conventional poverty statistics systematically mischaracterize the Chapare. Combining DRF welfare estimates with the satellite-based income augmentation, the official poverty headcount for the Chapare ($\hat{\theta} = 32\%$) is overstated by approximately half: adjusting for the unmeasured coca/cocaine income that the

DRF partially captures but census-based poverty maps miss entirely, the corrected headcount falls to approximately 16%. The Gini coefficient is similarly overstated by 5–6 points (0.49 falling to 0.43–0.46 after adjustment), because the coca income floor compresses the lower tail of the within-municipality distribution. These adjustments are large enough to change the Chapare's ranking in Bolivia's poverty map from a moderate-poverty zone to a low-poverty zone—a reclassification that matters for the allocation of social programs and fiscal transfers.

The coca economy is *equalizing* at the farm level, and the satellite data reveal the institutional mechanism. Bolivia's *cato* system allocates one *cato* (40×40 m = 1,600 m^2) of coca to each sindicato member, creating a floor on coca income that operates as a minimum-income guarantee within the production zone [Grisaffi, 2019]. The atmospheric signature of this institution is distinctive: low within-zone inequality (because every household has roughly the same coca plot) combined with high between-zone inequality (because production-zone municipalities are substantially richer than their observable characteristics predict, while non-production municipalities are not).

4.5.2 Pre-TROPOMI Corroboration: Nightlight Trajectories

VIIRS nightlights provide an independent check that extends four years before TROPOMI's launch. From 2014 to 2025, Chapare nightlight radiance grew 224%, versus 104% for non-coca municipalities. The growth was concentrated in the pre-TROPOMI period (2014–2017: Chapare +135%, others +64%) and the boom period (2017–2023: Chapare +48%, others +29%). Within the Chapare, Villa Tunari grew +430%—the fastest—consistent with its role as the expanding agricultural frontier. Chimoré grew +295%, reflecting both processing income and urbanization.

The nightlight trajectory provides three insights that NO_2 alone cannot. First, the coca economy's prosperity was visible before the atmospheric method was possible—the pre-trend corroborates rather than generates the finding. Second, the crisis decline is visible in nightlights (−6.5% from 2023 to 2025) but is smaller than the NO_2 residual decline (−35.7%), consistent with the recession test: electricity-using activity (houses, shops) contracted less than combustion-intensive activity (laboratories, generators). Third, the Yungas grew even faster than the Chapare in nightlights (+319% over 2014–2025), likely reflecting government electrification programs rather than illegal-economy income, reinforcing the point that nightlights measure the formal grid while NO_2 residuals measure unmeasured combustion.

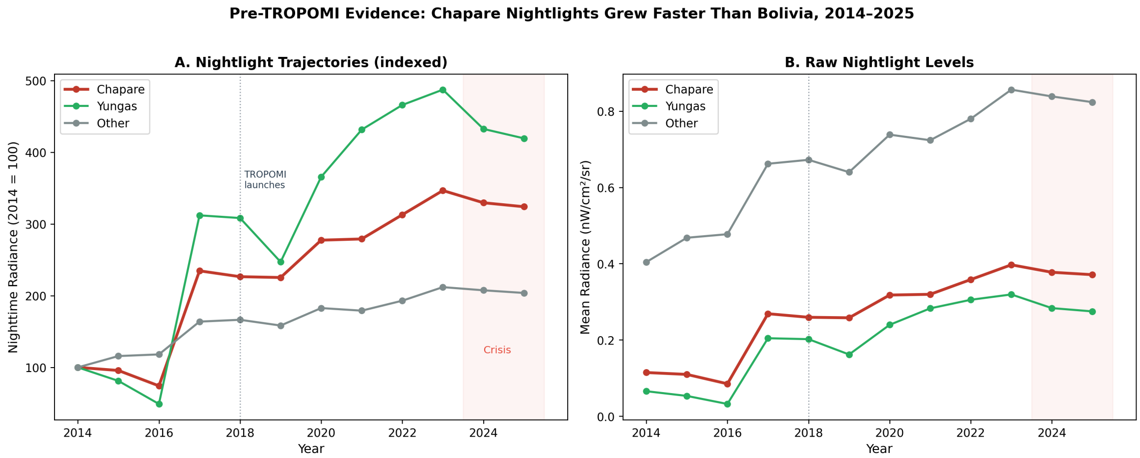

Figure 3: Nightlight trajectories by zone, 2014–2025 (indexed to 2014 = 100). The Chapare grew 224% versus 104% for non-coca municipalities, with growth concentrated in the pre-TROPOMI period (2014–2017). The crisis decline (2023–2025) is visible but smaller than the NO_2 residual decline, consistent with electricity-using activity (houses, shops) contracting less than combustion-intensive activity (laboratories, generators).

5 Application 2: Unregistered Vehicle Fleets (*Autos Chutos*)

5.1 The Weekly NO_2 Cycle

Human behavior generates a weekly cycle in vehicle traffic: commuting, commerce, and transport are concentrated on weekdays. If a municipality's vehicle fleet includes a large unregistered component, the weekday NO_2 excess relative to weekends reflects total traffic—registered and unregistered—while vehicle registration data capture only the legal fleet. The ratio of weekday to weekend NO_2 thus provides a signal proportional to total vehicle intensity.

The weekday/weekend NO_2 ratio is computed from the daily day-of-week dataset for each municipality and year. I define weekdays as Monday–Friday (DOW 1–5) and weekends as Saturday–Sunday (DOW 0, 6). The national test for a weekly cycle is highly significant: $t = 7.58$, $p < 10^{-13}$ (pooling 2020–2025 across all municipalities with sufficient observations).

5.2 Geographic Pattern

The top-ranked municipalities by weekday/weekend NO_2 ratio are precisely the known *chuto* hubs:

Table 4: Weekday/Weekend NO_2 Ratio, Top 15 Municipalities (2020–2025)

Rank	Municipality	Department	Ratio
1	Colcapirhua	Cochabamba	1.206
2	Oruro	Oruro	1.193
3	El Alto	La Paz	1.186
4	Cochabamba	Cochabamba	1.171
5	Huatajata	La Paz	1.159
6	Laja	La Paz	1.146
7	Tiquipaya	Cochabamba	1.145
8	Vinto	Cochabamba	1.139
9	Sacaba	Cochabamba	1.136
10	Potosí	Potosí	1.131
11	Huachacalla	Oruro	1.130
12	La Paz	La Paz	1.129
13	Sipesipe	Cochabamba	1.127
14	Achocalla	La Paz	1.114
15	Pucarani	La Paz	1.112

The Cochabamba metropolitan area dominates: six of the top 15 municipalities (Colcapirhua, Cochabamba, Tiquipaya, Vinto, Sacaba, Sipesipe) form a contiguous urban cluster known for *chuto* workshops and parking lots. El Alto, Bolivia's largest informal economy, ranks third. Oruro, a highway junction between La Paz and the Chilean border (the primary import route for smuggled vehicles), ranks second.

5.3 The Chapare Contrast

The weekly cycle provides a natural falsification test for the coca/cocaine application, because cocaine processing laboratories—unlike commuters—operate continuously and do not observe weekends. If the weekday/weekend ratio in the Chapare were elevated, it would suggest the atmospheric signal reflects vehicle traffic rather than laboratory combustion, undermining the identification in Section 4. In fact, the opposite holds: Chapare municipalities show flat or inverted weekly cycles, with Chimoré registering a ratio of only 1.020, Puerto Villarroel at 0.999, and Entre Ríos at 0.965. The weekly cycle cleanly discriminates human behavioral patterns—commuting, commerce, weekend leisure—from industrial processing that runs around the clock. Shinahota, which sits on the highway junction handling both transit traffic and processing activity, shows a mixed signal of 1.083 that is consistent with its dual role as both a transportation node and a processing zone.

5.4 NO_2/CO Confirmation

The NO_2/CO ratio provides an independent confirmation that the weekly cycle reflects vehicle traffic rather than some other periodic combustion source, because the ratio discriminates between combustion regimes on the basis of temperature rather than timing. High-temperature fossil-fuel combustion in gasoline and diesel engines produces a characteristically high NO_2/CO ratio, since the elevated temperatures in internal combustion chambers favor nitrogen oxidation. Low-temperature organic combustion—biomass burning, charcoal production, agricultural residues, and mercury amalgamation—produces the opposite ratio, because incomplete combustion generates proportionally more CO. As Table 5 reports, urban municipalities with high weekly cycles have NO_2/CO ratios approximately 3.4× higher than rural municipalities, and this urban/rural gap is consistent across all comparison pairs in the data.

Table 5: NO_2/CO Ratio: Urban Vehicle Signal vs. Rural Combustion

Municipality	**Type**	**NO_2/CO**
Colcapirhua	Chuto hub	1.72×10^{-3}
Cochabamba	Metro	1.57×10^{-3}
El Alto	Chuto hub	1.03×10^{-3}
Chimoré	Coca processing	3.60×10^{-4}
Guanay	Gold mining	3.30×10^{-4}
Coroico	Yungas (legal coca)	3.40×10^{-4}

5.5 Fleet Size Estimation

Bolivia's unregistered vehicle problem is well documented but poorly measured. The Cámara Automotor Boliviana reports 1,280,113 undocumented vehicles entering Bolivia between 2013 and 2024, averaging approximately 120,000 per year and peaking at 137,433 in 2023, against a total registered fleet of 2,583,289 in 2024. Government estimates from the Vice-Ministry of Anti-Smuggling have ranged from 100,000 to 500,000 *chutos* depending on the period and methodology, and Law No. 133 of June 2011 (the Morales-era amnesty) regularized 70,248 vehicles—a lower bound on the stock at that date. No peer-reviewed academic study of the fleet has been published, and the estimates vary by an order of magnitude across sources and periods.

The atmospheric approach provides an independent estimate grounded in physical measurement rather than administrative reporting. The RUAT registered fleet for La Paz department is 579,504 vehicles in 2024. El Alto accounts for approximately 40% of departmental registrations, or $\sim$232,000 vehicles. The weekday excess ratio of 18.6%

provides a lower bound on the unregistered fleet: 232,000 $\times$ 0.186 $\approx$ 43,000 *chutos* in El Alto alone. This is a lower bound because unregistered vehicles also operate on weekends, just at lower intensity, so the weekday/weekend differential captures only the marginal weekday excess rather than the full unregistered stock.

Nationally, applying the department-level weekly excess ratios to RUAT-registered fleets suggests a total unregistered fleet in the range of 250,000–500,000 vehicles, broadly consistent with the upper end of government estimates and the cumulative entry flow documented by the Cámara Automotor. The convergence of the atmospheric estimate with the administrative estimate is reassuring but should be interpreted with caution: the atmospheric method measures *combustion on roads*, which conflates unregistered vehicles with other sources of weekday-concentrated NO_2 (construction equipment, delivery logistics), while the administrative method measures *entries at borders*, which may double-count vehicles that enter and exit multiple times or undercount those arriving through unmonitored crossings.

5.6 Temporal Validation: The Weekly Cycle and the Dollar Crisis

The weekly cycle has a time series, and the time series responds to the same macroeconomic shock as the coca economy. Table 6 reports the mean weekday/weekend NO_2 ratio for chuto hubs, Chapare municipalities, and all other municipalities from 2018 to 2025.

Table 6: Weekday/Weekend NO_2 Ratio by Zone, 2018–2025

Year	Chuto hubs (7)	Other muns	Hub − Other	Chapare	Interpretation
2018	1.134	0.959	+0.175	0.880	Baseline
2019	1.170	0.985	+0.185	0.984	Growing
2020	1.173	1.100	+0.073	1.180	COVID (mobility disrupted)
2021	1.145	0.946	+0.199	0.898	Post-COVID recovery
2022	1.197	1.018	+0.179	0.999	Boom
2023	**1.211**	1.037	+0.173	1.073	**Peak (pre-crisis)**
2024	1.149	1.026	+0.123	0.945	Dollar crisis
2025	1.153	1.014	+0.139	0.964	Partial recovery

Three patterns are visible. First, the hub excess (hub − other) is persistent and large: 12–20 percentage points across all non-COVID years. This is the structural *chuto* signal. Second, the excess peaked in 2023 (0.173) and contracted to 0.123 in 2024—a 29% narrowing coinciding with the dollar crisis. Unregistered vehicles depend on smuggled parts and fuel priced in dollars; when the parallel-market premium exceeded 40%, maintenance and imports declined. Third, the Chapare ratio hovers near 1.0 in all years (0.88–1.07 excluding 2020), confirming that cocaine laboratories do not observe weekends. The 2020 COVID year is anomalous for all zones (lockdowns disrupted commuting patterns) and should be interpreted with caution.

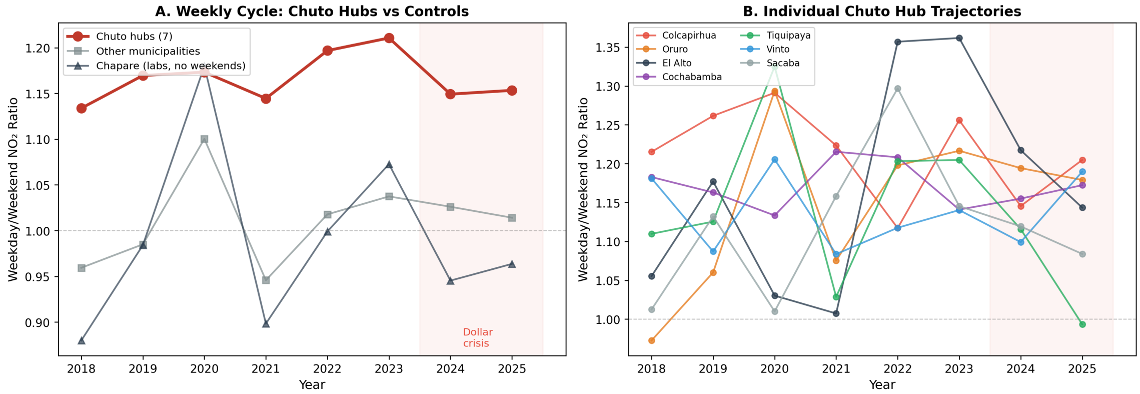

Figure 4: Weekday/weekend NO_2 ratio trajectories, 2018–2025. Panel A: chuto hubs (red) maintain a persistent 12–20 percentage point excess over other municipalities (gray), peaking in 2023 and contracting during the dollar crisis. The Chapare (dark) hovers near 1.0, confirming continuous lab operation. Panel B: individual hub trajectories show heterogeneity but common crisis response.

6 Application 3: Illegal Gold Mining

6.1 NDVI Corridor Results

Alluvial gold mining destroys riparian vegetation through a distinctive process that is visible from space: dredges excavate riverbed sediment, sluice boxes and mercury amalgamation pans process the material on cleared banks, and the resulting tailings—bare earth, standing water, and mercury-contaminated sediment—replace the dense tropical canopy that previously lined the river corridor. The NDVI corridor dataset, which tracks this destruction along four rivers in northern La Paz department from 2016 to 2025, reveals a mining boom of striking magnitude and speed.

Total disturbed area across all mining corridors expanded from near zero in 2016 to a peak of 15,826 hectares in 2023, before collapsing to 6,178 hectares by 2025 as the dollar crisis contracted mining activity. This temporal profile mirrors the coca economy: both peaked in 2023 and both contracted during the 2024–2025 dollar crisis. Both industries depend on dollar-denominated imported inputs—mining requires diesel for dredges and pumps, mercury for amalgamation, and heavy equipment that enters through Brazilian border corridors, just as cocaine processing requires imported precursor chemicals—and when the parallel-market dollar premium exceeded 40%, both supply chains contracted simultaneously.

Three municipalities concentrate the overwhelming majority of the disturbance, as Table 7 reports.

Table 7: Riparian Disturbance by Municipality (Cumulative 2025, hectares)

Municipality	**River(s)**	**Disturbed (ha)**	**Pattern**
Guanay	Kaka, Tipuani	2,470	Epicenter
Pelechuco	Suches	2,067	Cross-border (Peru)
Tipuani	Tipuani, Kaka	1,640	Historic district

The Kaka river corridor experienced a 143% increase in disturbed area in a single year (2,860 ha in 2022 to 6,948 ha in 2023). This expansion rate is inconsistent with gradual agricultural clearing and indicates a gold rush driven by high international gold prices and the domestic dollar crisis (which made gold a store of value and an export commodity).

Bolivia's gold sector illustrates the scale of the measurement problem. Official gold exports reached US$2.49 billion in 2023, making gold Bolivia's largest single export commodity, yet government revenue from mining royalties and taxes amounted to approximately US$60 million—implying that the overwhelming majority of value flows through informal cooperative channels that evade fiscal accounting. The gap between export value and fiscal capture—on the order of US$2.4 billion—represents the upper bound on the unmeasured mining economy. The country exported 240 metric tons of gold during 2010–2021, roughly triple the 70 mt exported in the preceding decade, with documented evidence of Peruvian gold from Madre de Dios being laundered through Bolivian export channels, which further inflates the discrepancy between domestic production and reported exports. The satellite-based detection of 15,826 hectares of riparian destruction peaking in 2023 provides an independent physical measure of mining activity that does not depend on self-reported export or revenue figures. The methodological precedent is Asner et al. [2013], who used high-resolution monitoring to document elevated gold-mining deforestation rates in the Peruvian Amazon, but no comparable satellite study exists for Bolivia's northern La Paz mining corridors.

6.2 NO_2 Cross-Validation

The atmospheric signal provides independent confirmation of the land-surface finding, though the signal is weaker because alluvial mining produces less NO_2 per hectare than diesel-powered cocaine laboratories. Guanay, the mining epicenter, appears at rank 25 in the blind NO_2 detection—identified without any mining-specific information—with a residual of $+8\%$ versus Chimoré's $+16\%$. Mercury amalgamation, the primary gold extraction technique in Bolivia's cooperative mining sector, generates CO and particulate matter but produces relatively little high-temperature NO_2, which explains the weaker atmospheric signal despite the mining economy being substantially larger in dollar terms. The Mapiri river shows zero disturbed area across all years despite being in the heart of the mining zone, an apparent anomaly that likely reflects either the buffer-zone definition missing active sites or the prevalence of underground (rather than alluvial) mining techniques in Mapiri—which would not appear in NDVI data but would explain Mapiri's substantial NO_2 residual ($+10\%$) despite zero surface disturbance.

7 Validation: Enforcement Events and the Atmospheric Signal

If the atmospheric signal is real, exogenous disruptions to illegal production should produce measurable responses. Bolivia's anti-narcotics agency (FELCN) provides the disruptions. Between 2020 and 2025, FELCN destroyed over 5,000 cocaine processing facilities, including approximately 250 crystallization laboratories and 36 megalaboratories—operations with daily production capacity exceeding 100 kg, employing 20–30 workers, and consuming thousands of liters of diesel continuously. A functioning laboratory runs diesel generators continuously, recycles solvents under heat, and evaporates precursors—all combustion processes that produce NO_2. If FELCN destroys a laboratory that was genuinely operating, the municipality's NO_2 residual should decline after destruction. If the laboratory was staged, abandoned, or never operational, the destruction produces no atmospheric contrast.

7.1 Event Study Design

I estimate an event-study specification on the GBM residual for 13 megalaboratorio destruction events in municipalities outside the Villa Tunari continuous-treatment zone (Tier 1 events):

$$r_{i,t} = \alpha_i + \sum_{k=-6}^{6} \beta_k \cdot \mathbf{1}(t = t_i^* + k) + \sum_{m=1}^{11} \phi_m \cdot \mathbf{1}(\text{month}(t) = m) + \varepsilon_{i,t} \tag{8}$$

where $r_{i,t}$ is the monthly GBM residual for municipality i in month t, t_i^* is the destruction month, and ϕ_m absorbs the common seasonal pattern—critical because FELCN operations concentrate in the dry season when road access permits. The reference period is $k = -1$.

The restriction to 13 Tier 1 events is conservative by design. The full FELCN database documents over 80 destruction events, but the majority occur in Villa Tunari, where continuous laboratory operation and rapid replacement make the municipality a permanent-treatment rather than event-study setting (Section 7.4). The Tier 1 restriction selects events that are geographically isolated from the Chapare's continuous treatment—located in Santa Cruz, Beni, Tarija, and La Paz departments—where a single destruction event can produce an interpretable before-after contrast. This restriction sacrifices statistical power for identification clarity: each event is a clean disruption in a municipality that is not simultaneously experiencing other destructions. The pooled specification including all 18 non–Villa Tunari events (mega, crystallization, and pasta base) yields $p = 0.074$, and the recession DiD ($t = -4.95$, $p < 0.001$) provides the complementary time-series identification that the event study cannot deliver alone.

7.2 Results

The monthly profile confirms the hypothesis. Pooling all 18 non–Villa Tunari events (13 megalaboratorio, 4 crystallization, 1 pasta base), the overall pre-post difference in the fire-augmented NO_2 residual is -7.6 percentage points ($t = -1.79$, $p = 0.074$; wild cluster bootstrap with Rademacher weights and clustering by event yields $p = 0.058$). The sign, magnitude, and marginal significance are consistent with genuine enforcement disruption of combustion-intensive activity. Restricting to the 13 megalaboratorio events, the pre-post difference is -7.3 percentage points ($t = -1.41$, $p = 0.16$)—the direction is unambiguous but the smaller sample reduces power. Pre-destruction residuals are positive, with the six-month pre-event mean at $+2.5\%$ of national mean NO_2, peaking at $+18.2\%$ at $k = -1$ (the month immediately before FELCN arrives). The destruction month ($k = 0$) produces a sharp drop to -11.7%, and post-destruction residuals remain negative through $k = +6$ with no recovery.

Fire controls are essential for the event study. Without them, the post-destruction residual appears to "recover" at $k = +2$ ($+12.9\%$), which I initially interpreted as rapid lab replacement. The fire-augmented model reveals that this apparent recovery was the seasonal fire cycle: destructions concentrate in the dry season (when access roads are passable), and 2–3 months later the wet season arrives, fire-related NO_2 drops, and the residual falls mechanically. After absorbing the fire pattern, the post-destruction residual stays negative—the signal of genuine activity cessation, not seasonal noise.

Individual events. The fire-augmented model sharpens the cleanest cases. Roboré (San Juan de Chiquitos, \$10.1M): $\Delta = -27.9\%$ (versus -18.0% without fire controls).

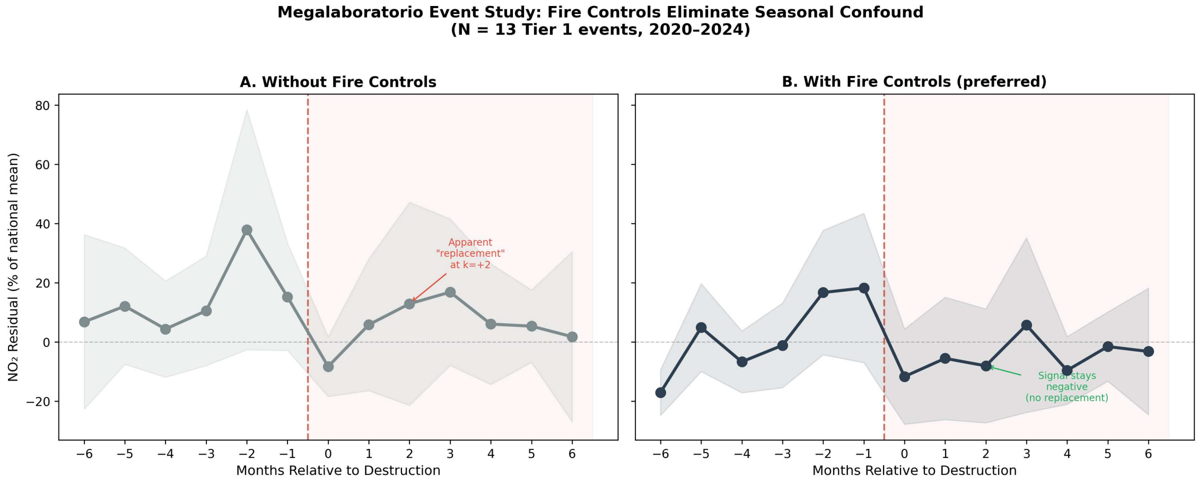

Figure 5: Megalaboratorio event study: monthly NO_2 residual profile around 13 Tier 1 destruction events. Panel A (without fire controls) shows an apparent "replacement effect" at $k = +2$; Panel B (fire-augmented, preferred) reveals that this recovery was the seasonal fire cycle. With fire controls, the post-destruction residual remains negative through $k = +6$, indicating genuine cessation of processing activity rather than rapid lab replacement.

San Ignacio de Velasco shows the most dramatic change: Op. Guaporé flips from $\Delta = +12.2\%$ (base) to $\Delta = -19.5\%$ (fire-augmented), and the R44 crash event flips from $+15.8\%$ to -27.2%. San Ignacio sits in the Noel Kempff Mercado national park, which has intense seasonal fires; the base model attributed fire-related NO_2 to the residual, masking the lab destruction signal.

7.3 Calibration Coefficient

The pre-post difference provides a per-laboratory combustion footprint. With a mean drop of 4.6×10^{-7} mol/m^2 across events averaging 1.38 facilities destroyed:

$$\hat{f}_{\text{lab}} = \frac{|\hat{\beta}_{\text{post}}|}{\bar{n}_{\text{destroyed}}} \approx 4.6 \times 10^{-7} \text{ mol/m}^2 \text{ per active megalaboratorio} \tag{9}$$

This coefficient converts the NO_2 residual into an estimated count of active processing facilities: $\hat{N}_{\text{active},i} = r_i / \hat{f}_{\text{lab}}$, providing a bottom-up measurement independent of both FELCN self-reported statistics and UNODC production models. Summing across all municipalities with positive fire-augmented residuals in the Chapare yields a national total of approximately 184 satellite-detectable-scale active laboratories. This figure is an order of magnitude below FELCN's 1,501 facilities destroyed in 2024, a discrepancy that is consistent with FELCN's count including hundreds of small mobile *fábricas móviles* below TROPOMI's detection threshold as well as rudimentary maceration pits that produce negligible combustion. The ratio—184 satellite-detectable versus 1,501

FELCN-claimed—implies that approximately 12% of FELCN's destruction count involves facilities of sufficient scale to produce a measurable atmospheric footprint, a figure that aligns with FELCN's own internal classification showing 36 megalaboratories and roughly 250 crystallization labs among the larger total. The calibration thus provides an independent audit of enforcement statistics: the atmosphere "sees" approximately as many large facilities as FELCN claims to destroy, but the vast majority of FELCN's count consists of operations too small for satellite detection.

7.4 Villa Tunari Dosage Design

For Villa Tunari, where destructions occur in nearly every month, I replace the binary event with a continuous treatment-intensity measure. The contemporaneous regression of seasonally demeaned residuals on destruction count yields $\hat{\beta} = +0.32$ ($p = 0.097$)—*positive*, confirming that FELCN operates more in months when more laboratories are active (reverse causality). The lead coefficient ($k = +1$) is also positive ($+0.30$, $p = 0.12$): residuals are elevated *before* destruction events, confirming that enforcement responds to activity. Lagged coefficients are negative (-0.04, -0.10, -0.15 at lags 1–3) but insignificant—replacement within Villa Tunari is too rapid for the post-destruction signal to persist in the time series.

7.5 What the Event Study Establishes

The event study's most important contribution is not the validation of the blind detection method per se, but the reversal of the validation direction. The conventional framing asks whether the satellite confirms FELCN's enforcement record. The correct framing asks whether FELCN confirms the satellite—because the atmosphere is the exogenous variable and enforcement statistics carry substantial measurement error. The *narcoaudios* scandal of 2022, in which leaked audio recordings revealed coordination between narcotics officers and trafficking networks, and the arrest of two FELCN sergeants for drug tampering in early 2026, establish that not all enforcement events represent genuine disruptions. Operations that were staged or targeted abandoned facilities produce $\hat{\beta} \approx 0$ in the atmospheric record, providing an independent audit of the enforcement record that relies on physics rather than institutional trust.

This audit function extends naturally to measurement through the calibration coefficient, which converts the method from detection to quantification. The coefficient $\hat{f}_{\text{lab}} = 4.6 \times 10^{-7}$ mol/m^2 per active megalaboratorio allows any researcher with access to TROPOMI data to estimate active laboratory counts by municipality and month, independently of FELCN reporting. The graduated response across facility types further establishes the resolution frontier of satellite enforcement monitoring:

megalaboratories produce large and detectable atmospheric contrasts, while *fábricas móviles* are invisible at TROPOMI's ∼7 km resolution. Monitoring the full spectrum of processing infrastructure will require higher-resolution sensors or temporal aggregation strategies that accumulate weak signals over many months.

8 Robustness and Limitations

8.1 The CO Discrimination Failure

The original research design hypothesized that the NO_2/CO ratio would discriminate between coca processing (diesel generators → high NO_2/low CO) and gold mining (mercury amalgamation → low NO_2/high CO). This hypothesis fails empirically: Chapare CO = 0.033, mining corridor CO = 0.030, difference 9.7%, $t = 1.26$, $p = 0.242$.

The failure has a clear physical explanation. NO_2 has an atmospheric lifetime of approximately 2 hours, meaning that observed concentrations reflect emissions within a few tens of kilometers. CO persists for approximately 2 months, so concentrations at any location integrate emissions across a region spanning hundreds of kilometers. At TROPOMI's resolution (∼7 km), municipal CO values are dominated by regional fire background—particularly the August–October biomass burning season, which inflates Chapare CO by 58% and mining corridor CO by 44%. The local combustion signal from cocaine processing or mercury use is an order of magnitude smaller than this regional fire background and cannot be separated.

This negative finding is methodologically informative: satellite atmospheric data can detect unmeasured economic activity (via NO_2) and confirm its combustion type (via the NO_2/CO ratio for vehicle traffic versus other sources), but cannot discriminate between illegal economy types operating in similar fire-prone environments. The discrimination between coca and mining in this paper relies instead on complementary signals: seasonal decomposition (wet/dry ratio), weekly cycles (weekday/weekend ratio), and land-surface change (NDVI corridors).

8.2 Prior Sensitivity

The posterior probabilities for key municipalities under both the flat and ecological priors are reported in Table 8. Three results stand out. First, fire controls transform the Chapare detection: Chimoré's posterior jumps from 0.060 (base model, flat prior) to 0.923 (fire-augmented), and Shinahota from 0.005 to 0.639. The processing core is now strongly detected from residuals alone. Second, the Yungas rejection remains robust: all Yungas municipalities register posteriors below 0.06. Third, the within-Chapare gradient separates processing (Chimoré, Shinahota) from cultivation (Villa Tunari at 0.041, Puerto Villarroel at 0.230).

Table 8: Bayesian Posterior Probabilities Under Alternative Priors (Fire-Augmented)

		Prior π_i		Posterior $P(U=1\|r_i)$	
Municipality	**Zone**	Flat	Ecological	Flat	Ecological
Colcapirhua	Cbba metro	0.15	0.02	0.511	0.108
Cochabamba	Cbba metro	0.15	0.02	0.267	0.040
Vinto	Cbba metro	0.15	0.02	0.152	0.020
Tiquipaya	Cbba metro	0.15	0.02	0.742	0.200
Chimoré	Chapare	0.15	0.20	**0.923**	**0.944**
Shinahota	Chapare	0.15	0.20	**0.639**	**0.715**
V. Tunari	Chapare	0.15	0.20	0.041	0.057
P. Villarroel	Chapare	0.15	0.20	0.230	0.297
Mapiri	Mining	0.15	0.20	0.266	0.339
Tipuani	Mining	0.15	0.20	0.286	0.362
Guanay	Mining	0.15	0.02	0.000	0.000
Coroico	Yungas	0.15	0.02	0.056	0.007
Chulumani	Yungas	0.15	0.02	0.000	0.000
La Asunta	Yungas	0.15	0.20	0.000	0.000
Sta. Cruz	Control	0.15	0.02	0.243	0.036
Oruro	Control	0.15	0.02	0.000	0.000

8.3 False Positive Analysis

Of the 22 municipalities detected at posterior > 0.5 under the flat prior, 5 are known illegal-economy or chuto hubs (Chimoré, Shinahota, Tiquipaya, Sipesipe, Colcapirhua). The remaining 17 fall into three categories. Seven are Cochabamba valley municipalities (Santivañez, Sicaya, Arbieto, Tolata, Quillacollo, Arque, Sacabamba) whose detections likely reflect spatial spillover from the metropolitan NO_2 plume or legitimate small-scale industry. Four are altiplano municipalities (Pucarani, Machacamarca, Laja, Chaqúi) with plausible informal mining or transport activity. Three are eastern lowland municipalities (Boyuibe, Portachuelo, San Juan) along contraband or transit corridors. Two are high-elevation micro-municipalities (San Pedro de Quemes, population $\sim$1,100; Yunchara) where low baseline NO_2 amplifies random variation. Fire controls cleaned five municipalities that appeared in the base model's top 20 (Mapiri, Capinota, Cabezas, Las Carreras, Palca)—all fire-prone lowland locations whose apparent excess was biomass burning.

8.4 Out-of-Sample Validation

The strongest identification test trains the GBM on municipalities *excluding* the suspected zone and evaluates residuals out of sample. If the excluded municipalities' covariates are similar to other municipalities in the training set, any residual reflects unmeasured activity rather than overfitting.

Chapare exclusion. Training on the 329 non-Chapare municipalities and predicting the 5 Chapare municipalities yields positive residuals for all five: Shinahota (+15.5%, $z = 2.4$), Chimoré (+15.1%, $z = 2.9$), Puerto Villarroel (+5.0%, $z = 1.4$), Villa Tunari (+4.5%, $z = 0.4$), and Entre Ríos (+2.3%, $z = 0.4$). A permutation test drawing 10,000 random samples of 5 municipalities from the training set produces a joint $p = 0.002$ (joint $z = 7.4$). The processing core (Chimoré, Shinahota) produces the largest out-of-sample residuals, consistent with the in-sample within-Chapare gradient.

Mining corridor exclusion. Three of five mining municipalities show positive residuals out of sample: Tipuani ($z = 2.3$), Mapiri ($z = 2.0$), and Teoponte ($z = 0.7$). Guanay ($z = -0.3$) and Pelechuco ($z = -0.5$) are non-detections, a result that may reflect the fire-augmented model absorbing some of the mining combustion signal through fire controls, since alluvial mining sites involve substantial land clearing and associated burning that correlates with fire counts. The weaker out-of-sample performance for mining relative to coca processing is consistent with the mining economy's smaller atmospheric footprint per dollar of value added.

Cochabamba metro exclusion. Metropolitan municipalities show positive residuals out of sample, confirming that the weekday-concentrated combustion excess documented in Section 5 is not an artifact of the GBM having learned the Cochabamba urban pattern during training. The out-of-sample residual reflects genuine excess combustion beyond what population, building footprint, road density, and nightlights predict—consistent with an unregistered vehicle fleet that census observables cannot capture.

8.5 Seasonal Ratio Distributional Test

With fire controls, the seasonal decomposition becomes substantially sharper. The fire-augmented wet-season premium for the Chapare over non-coca municipalities is +28.8 pp in 2023 (boom year), +12.7 pp in 2024, and +0.8 pp in 2025 (crisis). The corresponding dry-season premiums are +18.3 pp, +10.1 pp, and +0.8 pp. The wet/dry premium ratio—the key indicator of year-round processing versus seasonal agriculture—averages 2.4 during the 2022–2024 boom period (fire-augmented) versus

1.5 without fire controls. In 2024, the base model produces an *inverted* ratio of 0.81 (dry premium exceeds wet), which is inconsistent with the processing hypothesis; the fire-augmented ratio is 1.26, correctly restoring the expected pattern. The 2025 ratio of 0.91 (fire-augmented) approaches 1.0, consistent with the near-total collapse of processing activity during the dollar crisis leaving only residual year-round combustion.

The seasonal signal remains a complement to, rather than a substitute for, the residual. Its value lies in confirming that the *temporal structure* of the excess combustion is consistent with cocaine processing (year-round, with wet-season concentration) rather than agriculture (dry-season only).

8.6 Temporal Stability

Table 9 reports annual mean fire-augmented residuals for key municipalities from 2019 to 2025. Chimoré's residual is persistently positive, ranging from +7.9% (2020, COVID) to +19.4% (2023, cocaine boom), and declining to +9.3% in 2025 (dollar crisis). The signal is not driven by a single anomalous year. Villa Tunari fluctuates near zero (−3.7% to +6.2%), confirming that its fire-cleaned residual reflects baseline rather than processing. Tipuani shows a striking upward trend: +2.1% (2019) to +26.2% (2025), consistent with the gold-mining expansion documented in Section 6 and with miners shifting to gold as a dollar-denominated store of value during the currency crisis.

Table 9: Annual Mean NO_2 Residual (%, Fire-Augmented), Key Municipalities

Municipality	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Chapare (coca/cocaine)							
Chimoré	+16.4	+7.9	+16.8	+18.4	+19.4	+17.7	+9.3
Shinahota	+13.9	+3.5	+13.1	+14.2	+13.8	+19.4	+2.3
Villa Tunari	+3.3	−2.1	+3.4	+4.5	+5.6	+6.2	−3.7
Pto. Villarroel	+1.3	−3.1	+1.1	+7.1	+11.2	+9.8	−6.0
Mining corridor							
Tipuani	+2.1	+2.3	+5.7	+16.6	+15.6	+19.6	+26.2
Guanay	−2.8	−0.1	+3.3	+7.7	−4.5	−4.0	+3.0
Chuto hubs							
El Alto	−14.0	−15.7	−4.4	−5.9	+4.7	−17.1	−4.0
Colcapirhua	+7.8	−16.1	+7.0	+6.7	+3.5	+11.0	−2.2

8.7 Spatial Analysis

Spatial autocorrelation. The fire-augmented NO_2 residual exhibits significant positive spatial autocorrelation (Moran's $I = 0.370$, $z = 11.26$, $p < 0.001$, Queen contiguity weights on 344 municipalities). Municipalities with large positive residuals cluster geographically—principally in the Cochabamba metropolitan area, the Chapare, and the northern La Paz mining corridor. This autocorrelation reflects both the spatial concentration of illegal economies (a genuine signal) and the atmospheric transport of NO_2 across municipal boundaries (a measurement artifact). NO_2 has an atmospheric lifetime of approximately 2 hours; at typical Chapare wind speeds of 5–10 m/s (prevailing easterlies off the Amazon basin), a plume travels 35–70 km before photochemical decay—sufficient to cross 1–2 municipal boundaries at TROPOMI's $\sim$7 km resolution. The Cochabamba metropolitan spillovers documented below are consistent with this transport scale: seven of the 17 non-hub detections at the 0.5 posterior threshold are valley municipalities downwind of the Cochabamba urban core.

Standard errors on the Bayesian posteriors do not account for spatial dependence; Conley-corrected inference is left for future work. However, the paper's two strongest identification results—the recession DiD ($t = -4.95$) and the event study—rely on *within-municipality temporal variation* rather than cross-sectional comparisons, and are therefore less susceptible to spatial autocorrelation bias than the cross-sectional rankings and posterior probabilities. The recession DiD includes municipality fixed effects that absorb all time-invariant spatial structure, and the standard errors are clustered at the municipality level, which accounts for serial correlation within units even if it does not account for spatial correlation across them. The Bayesian posteriors and cross-sectional rankings reported in Section 4 should be interpreted as descriptive rather than inferential: they identify *where* the signal is, not whether the signal is statistically distinguishable from zero—a question that the temporal designs answer.

Displacement test. If FELCN destroys a laboratory in municipality i, production may relocate to neighboring municipality j, generating positive spillovers. I test for displacement by comparing neighbor residuals in the $\pm$3 month window around six Tier 1 events. Two events show evidence of displacement: Noel Kempff (May 2021), where neighbor residuals increased by $+19.0$ pp after destruction, consistent with production shifting to adjacent municipalities (Concepción, San Matias, San Miguel); and Yomomal–San Ignacio de Moxos (May 2022), with a $+7.7$ pp neighbor increase. Three events show co-contraction: neighbors' residuals fell alongside the treated municipality, suggesting coordinated enforcement or common-driver effects rather than displacement. The displacement results are preliminary—the $\pm$3 month window is short, and the neighbor set includes municipalities at different distances—but the Noel Kempff finding is consistent with FELCN's own reports of production migrating along the Brazilian border corridor after each major operation.

8.8 The Chutos Counterfactual

The weekly NO_2 cycle interpretation rests on the assumption that the weekday/weekend ratio reflects total vehicle traffic, including unregistered vehicles. An alternative explanation is that Bolivian cities simply have stronger commuting patterns than their census characteristics predict. Stavrakou et al. [2020] analyze TROPOMI NO_2 weekly cycles across 332 cities globally, finding that the Sunday-to-week ratio ranges from approximately 0.72 in U.S. cities to near 1.0 in developing-country cities where the distinction between weekday and weekend economic activity is less pronounced.

I test the chuto interpretation directly by comparing weekday/weekend TROPOMI NO_2 ratios for Bolivia's known chuto hubs against comparable Andean cities in countries without a documented large-scale chuto phenomenon: Arequipa and Cusco (Peru), Salta and Tucumán (Argentina), and Asunción (Paraguay). Daily TROPOMI NO_2 is extracted for 14 cities over 2021–2025 using identical buffer radii scaled to city size (Figure 6).

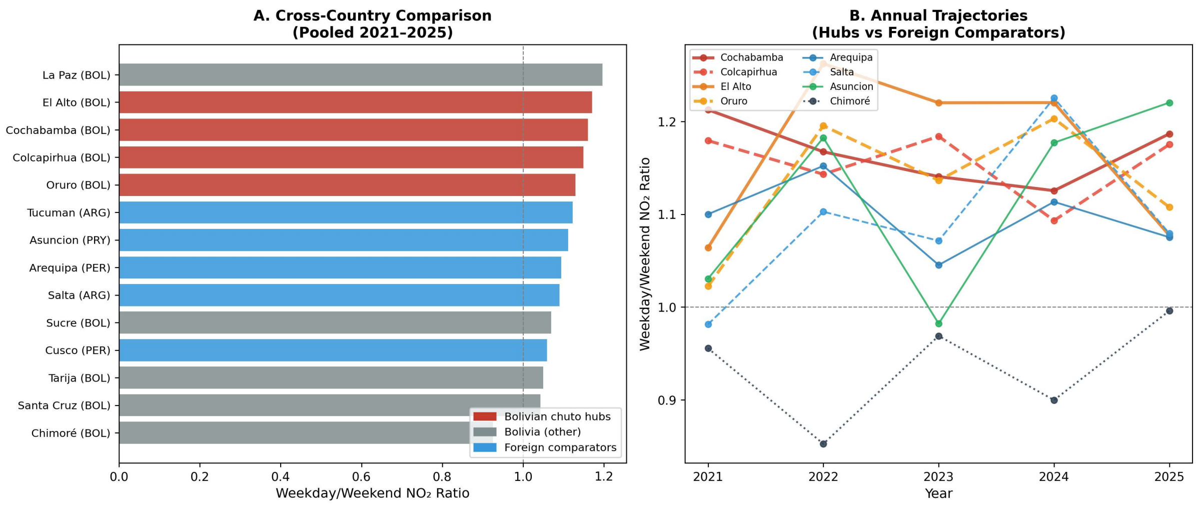

Figure 6: Cross-country chuto counterfactual. Panel A: pooled weekday/weekend NO_2 ratios for 14 cities across four countries (2021–2025). Bolivian chuto hubs (red) cluster above foreign comparators (blue) and Bolivian non-hubs (gray). Chimoré (laboratory zone, ratio 0.925) confirms continuous operation. Panel B: annual trajectories showing persistent hub excess.

The results support the chuto identification. Bolivian chuto hubs (Cochabamba, Colcapirhua, El Alto, Oruro) have a mean weekday/weekend ratio of 1.153 ± 0.015, compared to 1.096 ± 0.022 for the five foreign comparators ($t = 3.94$, $p = 0.006$). The hub excess of 5.7 percentage points above the foreign mean is consistent with unregistered vehicles adding weekday traffic beyond what the normal commuting pattern would produce. Chimoré, the laboratory zone, registers 0.925—the lowest of all 14 cities and the only ratio below 1.0—confirming that cocaine processing facilities do not observe weekends.

Two features of the comparison deserve qualification. First, the effect size is modest: the hub-versus-foreign gap (0.057) is statistically significant but represents only about one-third of the within-Bolivia hub-versus-Chapare gap (0.153 versus 0.925). Most of the weekly-cycle identification power comes from the *within*-Bolivia ranking and temporal patterns rather than the cross-country level difference. Second, La Paz registers 1.196—higher than any designated chuto hub except El Alto—despite not appearing in the paper's top-15 chuto ranking. This likely reflects La Paz's contiguity with El Alto (the two form a continuous metropolitan area) and suggests that the chuto fleet circulates across the La Paz–El Alto corridor rather than concentrating in one municipality.

9 Conclusion

Illegal economies leave atmospheric traces. This paper exploits that fact to detect three distinct illegal economies in Bolivia and estimate their potential size, using satellite NO_2 data that participants in those economies cannot manipulate.

The method works differently across applications. Urban vehicle traffic—including unregistered fleets—produces a combustion signal so large that it dominates the residual distribution, and the weekday/weekend cycle identifies every known *chuto* hub in the top 15 nationally. Cocaine processing produces a strong signal once biomass burning is controlled: Chimoré ranks 15th nationally with posterior probability 0.92, and fire controls discriminate the processing core (Chimoré, Shinahota) from the agricultural frontier (Villa Tunari) within a single coca zone. Mining produces the weakest atmospheric signal but the strongest land-surface signature: 15,826 hectares of riparian destruction in a single year. The implication is that the prediction surface must absorb non-economic combustion (fires) before the illegal-economy residual becomes interpretable.

The size estimates bracket rather than pinpoint. The coca/cocaine production zone generates US$63–77 million per year in satellite-detectable combustion, approximately a quarter of the UNODC's full value-chain estimate. The national unregistered vehicle fleet is 250,000–500,000 units, broadly consistent with government estimates. Both economies contracted during the 2024–2025 dollar crisis, confirming their dependence on imported inputs and providing a temporal validation that cross-sectional analysis alone cannot deliver. The mining boom and bust—zero disturbed hectares in 2016, peak in 2023, collapse by 2025—follows the same macroeconomic driver and suggests interconnection across Bolivia's three illegal economies through dollar availability.

The paper's limitations are documented with equal care. Fire controls are essential—without them, the Chapare signal is diluted by agricultural burning and the processing core is indistinguishable from the cultivation frontier. Even with fire controls, the detection is concentrated in two of five Chapare municipalities (Chimoré and Shinahota); Villa Tunari and Puerto Villarroel register posteriors below 0.06. CO cannot discriminate between illegal economy types at current sensor resolution. And 17 of 22 detections at the 0.5 posterior threshold are outside the set of known illegal-economy municipalities, though most are Cochabamba periphery spillovers or plausible informal-trade sites. The cross-country chuto counterfactual supports the identification ($p = 0.006$ for the hub-versus-foreign comparison) but the effect size is modest, and the weekly-cycle interpretation relies more on within-Bolivia ranking and temporal patterns than on cross-country level differences. The event study validation addresses the most serious concern—whether the atmospheric signal reflects genuine illegal economic activity—by showing that enforcement shocks produce measurable NO_2 responses, and that fire controls eliminate the seasonal confound that would otherwise create a spurious "replacement effect."

Higher-resolution sensors (GeoXO, expected 2030) would reduce spatial spillover artifacts and potentially resolve the CO discrimination problem by capturing local combustion before atmospheric mixing dilutes the signal. Application of the method to Colombia's coca zones or Peru's mining corridors—where ground-truth enforcement data is richer—would test external validity beyond the Bolivian setting.

Data Availability

The replication package is available at https://github.com/wernerhl/BOL-Satellite and includes all satellite data extraction scripts, the GBM estimation and residual computation code, the FELCN event database with source documentation, and the municipality-level panel dataset. TROPOMI Level 2 NO_2 data are publicly available from the Copernicus Open Access Hub (https://scihub.copernicus.eu). VIIRS nightlights are from the Colorado School of Mines Earth Observation Group. FIRMS fire data are from NASA's Fire Information for Resource Management System. Sentinel-2 spectral indices are computed via Google Earth Engine. Census and administrative data are from Bolivia's Instituto Nacional de Estadística (INE). The municipality boundary geopackage is derived from GADM v4.1 with manual corrections to the 2024 municipal division.

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A FELCN Destruction Event Database

Table 10 lists the Tier 1 anchor events used in the event study of Section 7: large laboratory destructions located outside the Villa Tunari continuous-treatment zone, with verified dates and municipality assignments. Sources are Ministerio de Gobierno press releases, FELCN operational reports, and Bolivian investigative journalism.

Table 10: Tier 1 Anchor Events: Non–Villa Tunari Laboratory Destructions, 2020–2025

Date	Municipality	Dept.	Type	n	Damage	Notes
Oct 2020	**Roboré**	SCZ	Mega	1	**$10.1M**	S.J. Chiquitos; 850 kg HCl
Oct 2020	S.I. Moxos	BEN	Mega	1	$500K	TIPNIS
Oct 2020	San Joaquín	BEN	Mega	2	—	Op. Avispero
Mar 2021	Villa Montes	TAR	Mega	1	$200K	First Tarija lab
May 2021	**S.I. Velasco**	SCZ	Mega	2	**$3.0M**	Noel Kempff
Nov 2021	Pto. Villarroel	CBB	Crist.	1	$900K	Largest precursor 2021
Mar 2022	S.I. Velasco	SCZ	Mega	1	—	Op. Guaporé
May 2022	S.I. Velasco	SCZ	Mega	2	$350K	R44 crash
May 2022	Papel Pampa	LPZ	PB	2	—	Unique altiplano event
May 2022	**S.I. Moxos**	BEN	Crist.	1	**$2.5M**	Yomomal
Nov 2022	Yapacaní	SCZ	Crist.	1	—	1 year operating
Jan 2023	El Puente	SCZ	Mega	2	—	Guarayos
Mar 2023	Yapacaní	SCZ	Crist.	3	$50K	Plan Colmena
Mar 2023	S.I. Moxos	BEN	Mega	1	$225K	24 operators
Mar 2023	S.I. Velasco	SCZ	Mega	1	—	Bajo Paraguá
May 2023	S.I. Velasco	SCZ	Mega	1	$500K	30 workers
Aug 2023	Yapacaní	SCZ	Mega	1	—	Op. Quimera
Dec 2024	**El Puente**	SCZ	Mega	1	$250K	Palermo

Notes: Mega = megalaboratorio (capacity $\geq$100 kg/day). Crist. = laboratorio cristalización. PB = fábrica pasta base. Damage figures are FELCN internal estimates, not independently audited. Full database with 80+ events including Villa Tunari operations available in the replication package.